Practice Problems
Calculus-Based Physics II

ame: Key **Date:** _____

Chapter V – Vector Review

1) A vector quantity has

(A) Both magnitude and direction.
B) Both mass and acceleration.
C) Both the action and reaction counterparts.
D) Direction only.
E) Magnitude only.

2) When a vector is drawn, what is indicated by the length of the vector?

A) The value of the quantity being measured.
(B) The magnitude of the quantity being measured.
C) The norm of the quantity being measured.
D) All the above.
E) None of the above.

3) What is the angle between parallel vectors?

A) 90°
(B) 0°
C) 180°
D) 270°
E) 45°

4) What must be equal for two vectors to be equal?

A) Magnitude (with unit) only
B) Direction only
(C) Both magnitude (with unit) and direction

5) If a vector is multiplied by 3 what happens to the vector's magnitude and direction?

(A) The vectors magnitude is tripled, and the direction is unchanged.
B) The vectors magnitude is divided by 3 and the direction is unchanged.
C) The vectors magnitude is tripled, and the direction is also tripled.
D) The vectors magnitude is unchanged, and the direction is tripled.

6) Which angle below would give a negative Dot product of two vectors? Why?

A) 90°
B) 0°
C) 280°
D) 45°
(E) 110°

$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos(\varphi)$

7) If the Dot Product between two vectors is equal to zero, what is the angle (ϕ) between the vectors?

A) 0°
B) 90°
C) 180°
D) 270°
E) 45°
(F) It can be either 90° or 270 °
G) It can be either 0° or 180 °

$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos(\phi)$

8) Two vectors have a given magnitude $|\vec{\mathbf{A}}| = 20$ N and vector $|\vec{\mathbf{B}}| = 5$ m calculate the Dot product for each orientation of vectors.

A) Vector $\vec{\mathbf{A}}$ is along the same direction as vector $\vec{\mathbf{B}}$? $\phi = 0°$

$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\phi = +100 \text{ N m}$

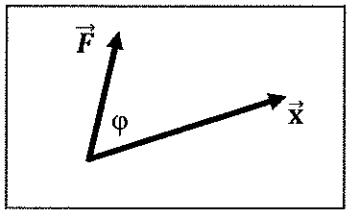
B) Vector $\vec{\mathbf{A}}$ is perpendicular to vector $\vec{\mathbf{B}}$?

$\phi = 90°$ $\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\phi = 0$

C) Vector $\vec{\mathbf{A}}$ is in the opposite direction to vector $\vec{\mathbf{B}}$?

$\phi = 180°$ $\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\phi = -100 \text{ N m}$

9) A force vector $\vec{\mathbf{F}}$ acts on a box with a magnitude of 400 N and the box moves with a displacement vector $\vec{\mathbf{x}}$ that has a magnitude of 10 m, the angle between the force and displacement is $\varphi = 30°$, calculate the work done on the box by the force by taking the Dot product of the two vectors? $W = \vec{\mathbf{F}} \cdot \vec{\mathbf{x}}$



$W = \vec{F} \cdot \vec{x} = |\vec{F}||\vec{x}| \cos\phi = 3,464 \text{ N m}$

10) The scalar component of a vector can best be described as

A) the direction of the vector.
B) the projection of the vector along a principle axis.
C) the magnitude of the vector.
D) the unit vector.

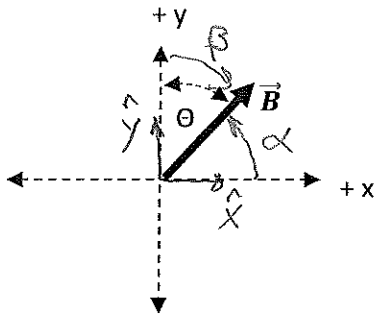
11) Given the vector below with a magnitude of 10 m and the angle $\Theta = 25°$, determine the angles α and β along with the components of the vector along x and y axis.

$\alpha =$ $90° - \Theta = 65°$

$\beta =$ $\Theta = 25°$

$B_x =$ $\vec{B} \cdot \hat{x} = |\vec{B}|\cos(\alpha) = 4.23\,m$

$B_y =$ $\vec{B} \cdot \hat{y} = |\vec{B}|\cos(\beta) = 9.06\,m$



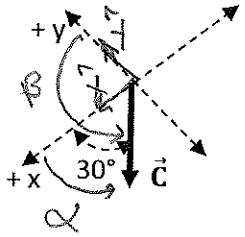
12) For vector $\vec{C}$ shown, having a magnitude of 120 N determine the angles α and β along with the components of the vector along the x and y axis? (Be careful of axis direction!)

$\alpha =$ $30°$

$\beta =$ $90° + 30° = 120°$

$C_x =$ $\vec{C} \cdot \hat{x} = |\vec{C}|\cos(\alpha) = 104\,N$

$C_y =$ $\vec{C} \cdot \hat{y} = |\vec{C}|\cos(\beta) = -60\,N$



13) Calculate the vectors magnitude and direction if the components of the velocity vector are $v_x = 5.6$ m/s and $v_y = 8.2$ m/s?

$\vec{V} = (V_x)\hat{x} + (V_y)\hat{y} = (5.6\,m/s)\hat{x} + (8.2\,m/s)\hat{y}$

$|\vec{V}| = \sqrt{V_x^2 + V_y^2} = 9.93\,m/s$

$\alpha = \cos^{-1}\left(\frac{V_x}{|\vec{V}|}\right) = 56°$ $\beta = \cos^{-1}\left(\frac{V_y}{|\vec{V}|}\right) = 34°$

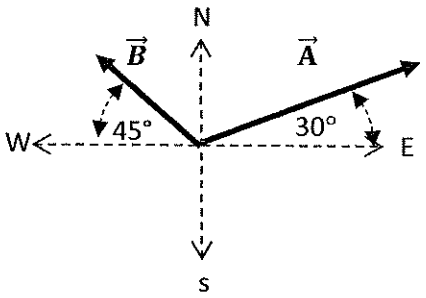
14) Vector $\vec{\mathbf{H}}$ is given in component form as $\vec{\mathbf{H}} = (5\ m)\hat{\mathbf{x}} + (-10\ m)\hat{\mathbf{y}}$, calculate the magnitude and direction (α and β) of the vector.

$|\vec{H}| = \sqrt{H_x^2 + H_y^2} = 11.18\ m$

$\alpha = \cos^{-1}\left(\frac{5m}{11.18m}\right) = 63°$

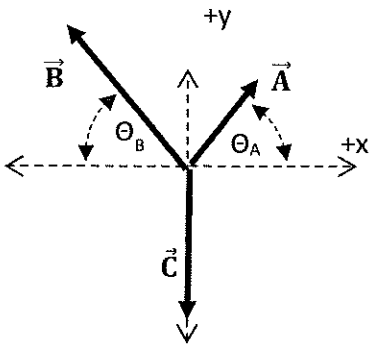
$\beta = \cos^{-1}\left(\frac{-10m}{11.18m}\right) = 153°$

15) Vector $\vec{\mathbf{A}}$ = 6.0 m and points 30° north of east. Vector $\vec{\mathbf{B}}$ = 4.0 m and points 45° north of west. Determine the resultant vector's $\vec{\mathbf{R}} = \vec{\mathbf{A}} + \vec{\mathbf{B}}$ direction (α and β).



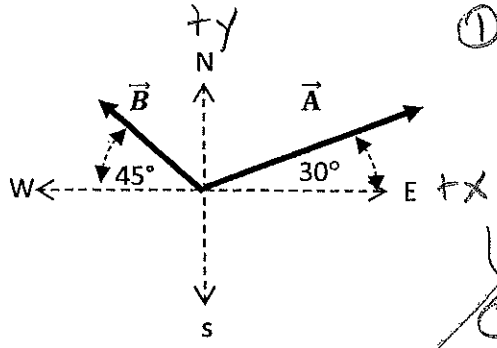
See attached

16) There are three force vectors acting on a box as shown below if vector $|\vec{\mathbf{A}}|$ = 490 N and Θ_A = 60°, Vector $|\vec{\mathbf{B}}|$ = 602.5 N and Θ_B = 66° above the negative x-axis, and vector $|\vec{\mathbf{C}}|$ = 669 N and is along the negative y-axis. What is the magnitude and direction (α and β) of the resultant forces $\vec{\mathbf{R}} = \vec{\mathbf{A}} + \vec{\mathbf{B}} + \vec{\mathbf{C}}$



See attached

15) Vector $\vec{A}$ = 6.0 m and points 30° north of east. Vector $\vec{B}$ = 4.0 m and points 45° north of west. Determine the resultant vector's $\vec{R} = \vec{A} + \vec{B}$ direction (α and β).



$\alpha_A = 30°$

$\beta_A = 90° - 30° = 60°$

$\alpha_B = 180° - 45° = 135°$

$\beta_B = 90° - 45° = 45°$

① $R_x = \vec{R}\cdot\hat{x} = \vec{A}\cdot\hat{x} + \vec{B}\cdot\hat{x}$

$= |\vec{A}|\cos(\alpha_A) + |\vec{B}|\cos(\alpha_B)$

$R_x = 5.2m + -2.8m = 2.4m$

② $R_y = \vec{R}\cdot\hat{y} = \vec{A}\cdot\hat{y} + \vec{B}\cdot\hat{y}$

$R_y = |\vec{A}|\cos(\beta_A) + |\vec{B}|\cos(\beta_B)$

$R_y = 3m + 2.8m = 5.8m$

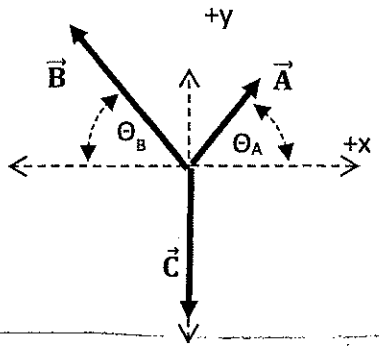
③ $\vec{R} = (2.4m)\hat{x} + (5.8m)\hat{y}$

$|\vec{R}| = \sqrt{R_x^2 + R_y^2} = 6.28m$

$\alpha = \cos^{-1}\left(\frac{R_x}{|\vec{R}|}\right) = 67°$

$\beta = \cos^{-1}\left(\frac{R_y}{|\vec{R}|}\right) = 23°$

16) There are three force vectors acting on a box as shown below if vector $|\vec{A}|$ = 490 N and $\Theta_A = 60°$, Vector $|\vec{B}|$ = 602.5 N and $\Theta_B = 66°$ above the negative x-axis, and vector $|\vec{C}|$ = 669 N and is along the negative y-axis. What is the magnitude and direction (α and β) of the resultant forces $\vec{R} = \vec{A} + \vec{B} + \vec{C}$



$\alpha_A = 60°$

$\beta_A = 90° - 60° = 30°$

$\alpha_B = 180° - 66° = 114°$

$\beta_B = 90° - 66° = 24°$

$\alpha_C = 90°$

$\beta_C = 180°$

$R_x = \vec{R}\cdot\hat{x} = \vec{A}\cdot\hat{x} + \vec{B}\cdot\hat{x} + \vec{C}\cdot\hat{x}$

$= |\vec{A}|\cos(\alpha_A) + |\vec{B}|\cos(\alpha_B) + |\vec{C}|\cos(\alpha_C)$

$R_x = 245N + (-245N) + 0 = 0N$

$R_y = \vec{R}\cdot\hat{y} = \vec{A}\cdot\hat{y} + \vec{B}\cdot\hat{y} + \vec{C}\cdot\hat{y}$

$= |\vec{A}|\cos(\beta_A) + |\vec{B}|\cos(\beta_B) + |\vec{C}|\cos(\beta_C)$

$R_y = 424N + 550N + -669N$

$R_y = 305N$

$\vec{R} = (0N)\hat{x} + (305N)\hat{y}$

$|\vec{R}| = \sqrt{R_x^2 + R_y^2} = 305N$

$\alpha = 90°$ $\beta = 0°$

17) Given the displacement vector of a bird with a time dependance below.

$$\vec{x} = ([-0.3\ m/s^2]\,t^2)\hat{x} + (5\ m/s + [4\,m/s]t)\hat{y}$$

Determine the magnitude and direction of the velocity of the bird after t = 10s of flight.

$\vec{v} = \frac{d\vec{x}}{dt} = \frac{d}{dt}\left[(-0.3\ m/s^2)t^2)\hat{x} + (5\ m/s + (4\ m/s)t)\hat{y}\right]$

$\vec{v} = (-0.6\ m/s^2 t)\hat{x} + (4\ m/s)\hat{y}$

$\vec{v}(10s) = (-6\ m/s)\hat{x} + (4\ m/s)\hat{y}$

$\alpha = \cos^{-1}\left(\frac{-6\ m/s}{7.2\ m/s}\right) = 146°$

$|\vec{v}| = \sqrt{v_x^2 + v_y^2} = 7.2\ m/s$

$\beta = \cos^{-1}\left(\frac{4\ m/s}{7.2\ m/s}\right) = 56°$

18) Given the velocity vector of a drone with a time dependance below, where b, c and d are constants

$$\vec{v} = ([bt^3)\hat{x} + (c + [4d]t)\hat{y}$$

A) Determine the units of the constants b, c and d. need units of m/s

Units of b: $bt^3 \rightarrow b(t^3) \rightarrow \left(\frac{m}{s^4}\right)t^3 \rightarrow m/s \rightarrow b \rightarrow m/s^4$

Units of c: $c \rightarrow m/s$

Units of d; $d \rightarrow dt \rightarrow \left(m/s^2\right)s \rightarrow m/s \rightarrow d \rightarrow m/s^2$

B) Determine the displacement vector in component form for the time interval of $t_i = 0$ m/s to $t_f = 4$ s.

$$\vec{x} = \int_{t_i}^{t_f} \vec{v}\,dt = \int_0^{4s} \left[(bt^3)\hat{x} + (c + 4dt)\hat{y}\right]dt$$

$$\vec{x} = \left[\left(\frac{bt^4}{4}\right)\hat{x} + \left(ct + 4\frac{dt^2}{2}\right)\hat{y}\right]\Bigg|_0^{4s}$$

$$\vec{x} = \left(\frac{b(256)}{4}\right)\hat{x} + (4c + 32d)\hat{y}$$

$$\vec{x} = (64b)\hat{x} + (4c + 32d)\hat{y}$$

Answers

1) A
2) B
3) B
) C
5) A
6) E
7) F
8) A) + 100 Nm B) 0 Nm C) - 100 Nm
9) +3,464 Nm
10) B
11) $\alpha = 65°$ $\beta = 25°$ $B_x = 4.23$ m $B_y = 9.06$ m
12) $\alpha = 30°$ $\beta = 120°$ $C_x = 104$ N $C_y = -60$ N
13) $|\vec{v}| = 9.93$ m/s $\alpha = 56°$ and $\beta = 34°$
14) $|\vec{H}| = 11.2$ m $\alpha = 63°$ and $\beta = 153°$
15) $|\vec{R}| = 6.28$ m $\alpha = 67°$ and $\beta = 23°$
16) $|\vec{R}| = 305$ N $\alpha = 90°$ and $\beta = 0°$
17) $|\vec{v}| = 7.2$ m/s $\alpha = 146°$ and $\beta = 56°$
18) A) b → m/s^4 c → m/s d → m/s^2 B) $\vec{x} = (64b)\hat{x} + (4c + 32d)\hat{y}$